

#ApexGP Racing

• Database & Power BI Analytics •



+



Power BI



MySQL
Database



Advanced
SQL Queries



Data Analysis
& Insights



Power BI
Dashboards



Security &
Optimization



PROJECT REPORT

*A Comprehensive Analysis of Formula-Style
Racing Championship Data*

Submitted By

Satyajit Pradhan



Table of Contents

1. Executive Summary..... 3

2. Project Background & Scenario..... 3

3. Database Architecture..... 3

4. SQL Implementation & Analysis.....5

5. Power BI Dashboard.....7

6. Key Insights..... 8

7. Tools & Technologies..... 8

8. Conclusion & Learning Outcomes..... 9

9. Author..... 9

1. Executive Summary

The ApexGP Racing Database System is an end-to-end data project built around a fictional Formula-style racing championship, the Apex Grand Prix. The project spans the complete analytics lifecycle: relational database design, advanced SQL implementation, and business intelligence dashboarding in Power BI.

The database models a global racing league in which ten teams field forty drivers across a season of races held at twenty circuits worldwide. Every practice, qualifying, and race lap is timed and logged, producing a dataset of roughly 30,000 records concentrated mainly in lap-by-lap timing data. This report documents the database architecture, the analytical SQL work performed against it, and the resulting Power BI dashboards used to surface championship insights.

The project was designed to mirror the kind of data challenges faced by real motorsport organisations — race engineers and strategists who need reliable, fast, and secure access to timing data in order to make split-second decisions.

2. Project Background & Scenario

The Apex Grand Prix is framed as a global open-wheel racing championship contested across a varied calendar of circuits — deserts, coastlines, mountains, and city streets. Ten constructors compete, each fielding four drivers, and every lap of every session is timed to the thousandth of a second by trackside sensors.

This timing data feeds a live database that is used in the scenario by race engineers, team strategists, the broadcast production crew, and the league's commercial and technical departments. Because pit strategy calls and championship outcomes hinge on split-second differences in lap time, the underlying database needs to support fast, accurate, and secure querying at scale.

To bring this scenario to life, 75 realistic operational situations were worked through — covering everything from identifying underperforming drivers and correcting mis-logged data, to building reusable views for broadcast graphics and setting up tiered database access for different roles. Rather than being pre-labelled by SQL technique, each situation had to be interpreted first, then solved — reflecting how real analytical requests arrive in practice.

3. Database Architecture

The database is built around five interconnected tables, designed to reflect the natural structure of a racing championship: constructors, drivers, venues, events, and the granular lap-by-lap timing data generated at each event.

3.1 Entity-Relationship Diagram

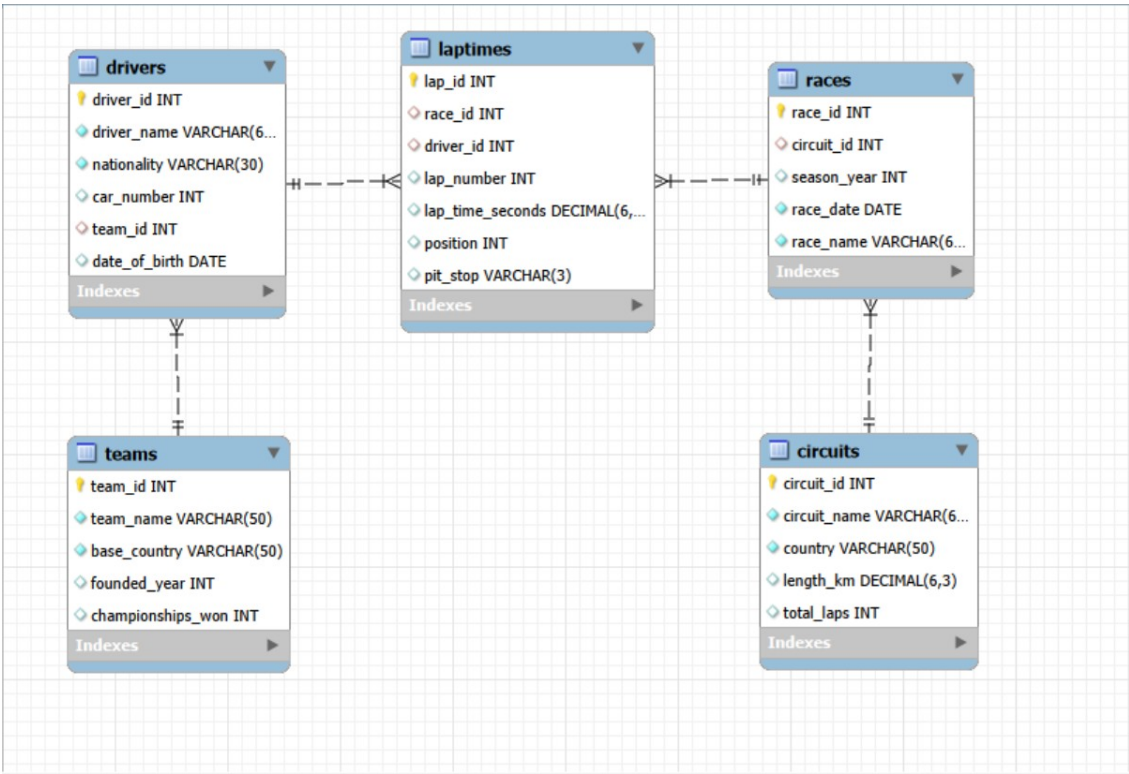


Figure 1 — Entity-Relationship diagram showing the five core tables and their relationships.

3.2 Table Overview

Table	Purpose	Key Fields
teams	Every constructor competing in the championship	team_id (PK), team_name, base_country, founded_year, championships_won
drivers	Every driver on the grid, linked to their team	driver_id (PK), driver_name, nationality, car_number, team_id (FK), date_of_birth
circuits	Every track on the calendar	circuit_id (PK), circuit_name, country, length_km, total_laps
races	Every individual race held at a circuit, across seasons	race_id (PK), circuit_id (FK), season_year, race_date, race_name
laptimes	Every single lap recorded by every driver in every race	lap_id (PK), race_id (FK), driver_id (FK), lap_number, lap_time_seconds, position, pit_stop

Relationships follow standard one-to-many patterns: a team has many drivers, a circuit hosts many races, and each race generates many lap-time records contributed by many drivers. At full scale the database holds

approximately 30,000 records, the overwhelming majority of which are individual lap times — a single race weekend alone generates thousands.

4. SQL Implementation & Analysis

Working against this schema, 75 real-world analytical and operational scenarios were solved, exercising the full breadth of SQL — from basic data retrieval through to advanced analytical and administrative techniques. The scenarios were grouped into six broad categories of work:

4.1 Analytical Querying

- Ranking and windowed comparisons — e.g. comparing each driver's lap time to their previous and next lap using window functions, to spot sudden pace changes.
- Aggregation and grouping — e.g. average lap time per circuit, pit stops per team, and driver counts per team.
- Multi-step (CTE-based) analysis — e.g. computing per-circuit average lap time first, then using it to flag circuits running slower than the season-wide norm.
- Segmentation via CASE logic — e.g. classifying laps as Fast / Average / Slow, or teams as Veteran / Established / New Entrant.

4.2 Data Integrity & Governance

- Constraints to prevent invalid data — e.g. rejecting negative lap times, rejecting finishing positions outside the number of cars on track, and enforcing that every driver belongs to a team.
- Referential integrity checks — e.g. identifying lap records that point to a driver ID no longer in the system, or laps logged against races that were never scheduled.
- Data cleaning — e.g. standardising inconsistent name capitalisation and trimming stray whitespace from circuit names.

4.3 Reporting Objects

- Views for recurring reporting needs — e.g. a live snapshot of each driver's name, team, car number and latest lap time for broadcast graphics, and a standing team/base-country/championship list for the commercial team.
- Consolidated summary queries — e.g. an end-of-season report combining roster size, average lap time, total pit stops, and an underperformance flag for every team.

4.4 Performance Optimisation

- Indexing strategy — e.g. indexing driver_id for fast lookups of a driver's full season history, and composite indexing on (race_id, lap_number) for the live timing dashboard's most common combined filter.
- Auditing existing indexes on the lap-times table after multiple seasons of ad-hoc changes.

4.5 Security & Access Control

- Role-based privilege management — e.g. read-only access for a new strategy analyst, elevated update rights for the chief timing officer, and full revocation of access for an outgoing freelance analyst.

4.6 Data Maintenance

- Bulk corrections — e.g. re-assigning an entire race's laps to the correct circuit in one operation, and applying a timing-offset correction across all laps at a single circuit.
- Archival and cleanup — e.g. identifying races older than two years for archiving, and purging obsolete test-session lap data.

5. Power BI Dashboard

The database was connected to Power BI Desktop to build a two-page interactive Championship Dashboard, translating the underlying SQL tables into a visual, filterable reporting tool for coaches, analysts, and commentators.

5.1 Championship Dashboard (Page 1)

The landing page gives a season-level view of the championship, with slicers for Season, Team, and Circuit.

Total Teams	Drivers	Races	Circuits	Avg Lap	Avg Position
10	40	50	20	93.9s	10.5

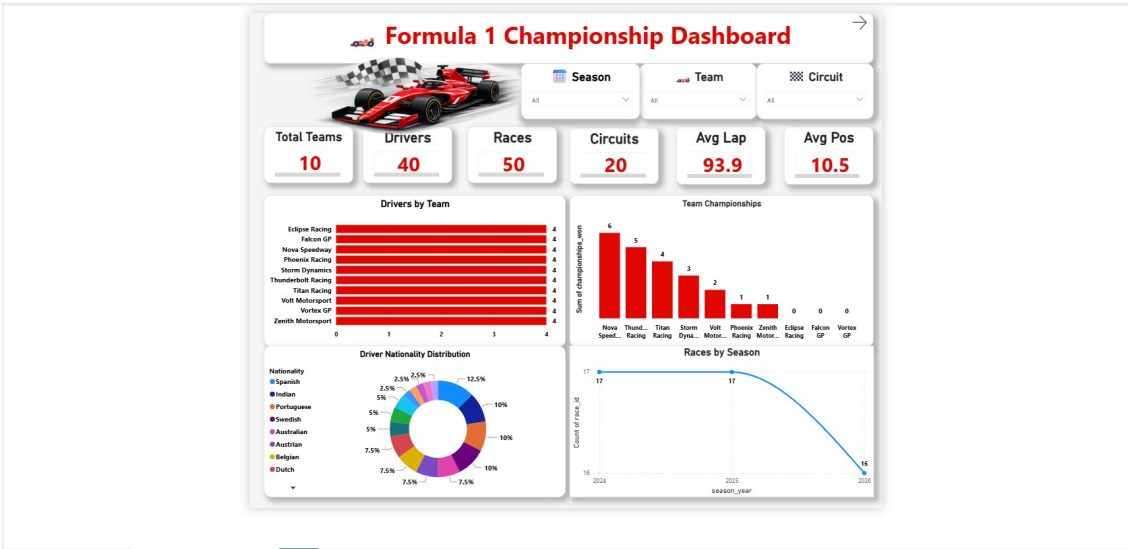


Figure 2 — Championship Dashboard: team standings, driver distribution, and season trends.

Key visuals on this page include a breakdown of drivers per team (evenly split at four per team), a ranked bar chart of team championships won — led by Nova Speedway with six titles, followed by Thunderbolt Racing (5) and Titan Racing (4) — a nationality breakdown of the driver grid, and a season-over-season race count showing 17 races in both 2024 and 2025, dropping to 16 in the current (2026) season.

5.2 Driver Performance Dashboard (Page 2)

The second page drills into individual driver performance, filterable by Season, Team, and Nationality.

Fastest Lap	Avg Lap Time	Avg Finish Position	Best Finish Position
78.00s	93.9s	10.50	1

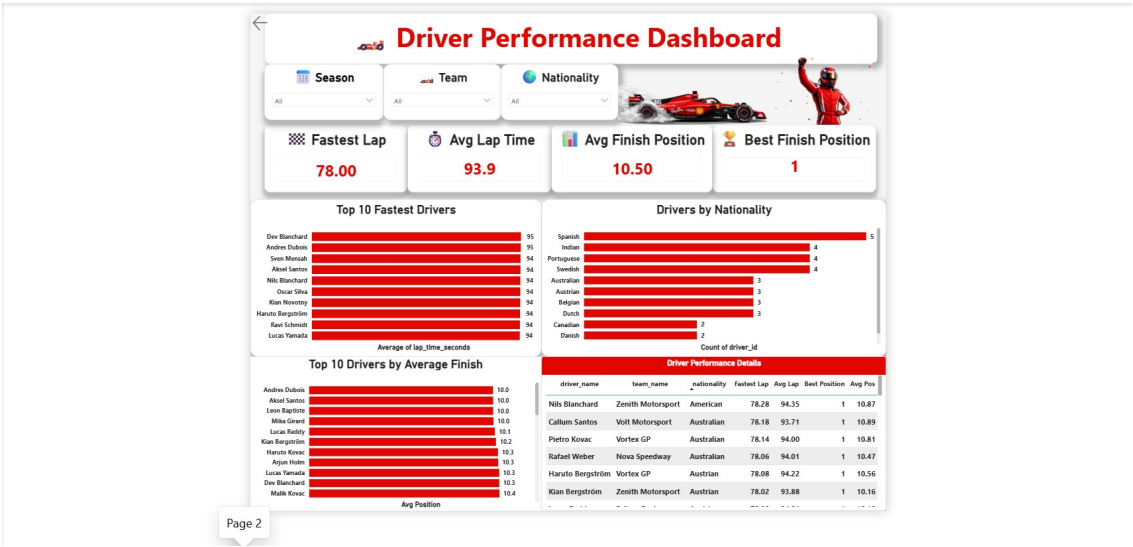


Figure 3 — Driver Performance Dashboard: fastest drivers, nationality split, and a full driver detail table.

This page ranks the ten fastest drivers by average lap time, breaks the grid down by nationality (Spanish drivers form the largest group), ranks drivers by average finishing position, and provides a detailed sortable table of every driver's fastest lap, average lap, best finish, and average position — giving analysts a single place to compare performance across the grid.

6. Key Insights

- Championship success is concentrated at the top: Nova Speedway, Thunderbolt Racing, and Titan Racing account for the majority of championships won, while three teams (Eclipse Racing, Falcon GP, Vortex GP) have yet to win a title.
- The driver grid is evenly structured — every team fields exactly four drivers — but performance and nationality are far more varied, with Spanish drivers forming the single largest nationality group.
- The season calendar has been broadly stable at 17 races per year, with a slight reduction to 16 races in the current season worth monitoring for future scheduling.
- The gap between the fastest recorded lap (78.00s) and the grid-wide average (93.9s) highlights the wide performance spread across circuits and driver skill levels, reinforcing the value of per-circuit and per-driver benchmarking rather than relying on a single season-wide average.

7. Tools & Technologies

Category	Tool
Database Engine	MySQL 8
Database Administration	MySQL Workbench
ER Diagramming	DBML / DrawDB
Business Intelligence	Power BI Desktop
Version Control	Git & GitHub

8. Conclusion & Learning Outcomes

This project provided end-to-end, hands-on experience across the data lifecycle — from designing a normalised relational schema, through writing advanced, real-world SQL across 75 distinct scenarios, to presenting the results as an interactive Power BI dashboard.

Working through the 75 scenarios reinforced practical skills in relational database design, query optimisation, analytical SQL (including CTEs and window functions), data integrity enforcement, database security and access management, and — critically — translating raw operational data into dashboards that a non-technical audience can act on. Together, these skills reflect the full workflow expected of a data analyst working with production-grade relational data.